

Michel Saliba

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EDUCATION

École Polytechnique Fédérale de Lausanne

Oct 2023 - Current

PhD Candidate in Nuclear Engineering

- My PhD project focalize around designing, building and initiating a new neutron imaging stating at the CROCUS zero-power reactor.
- Strong knowledge and know-how in diverse Monte-Carlo codes.
- Strong knowledge in neutron imaging, gamma spectroscopy, dosimetry, scatter camera systems, neutron noise analysis, nuclear security and safeguards.
- I1 - Radiation Protection Expert for Type B or C Work Sectors. Certified by the federal office of public health.
- Licensed Research Reactor Operator

ETH Zürich - École Polytechnique Fédérale de Lausanne

Sep 2021 - Sep 2023

Master of Science EPF-ETH in Nuclear Engineering

- Strong knowledge and know-how in Monte-Carlo neutron transport codes.
- Strong knowledge in Nuclear Materials, thermo-hydraulics, deterministic codes and nuclear reactor physics.

Univeristé Saint-Joseph de Beyrouth

Sep 2019 - Jun 2021

Bachelor in Physics

- Earned the distinction of being the top-ranked student in my class.
- Fundamentals in mathematics and physics.

Lebanese University Faculty of Engineering

Sep 2017 - Jun 2021

Three years of Mechanical Engineering

- Accomplished several optimisation projects.
- Fundamentals in programming and key engineering concepts.

EXPERIENCE

Kernkraftwerk Gösgen-Switzerland

Apr 2022 - Dec 2022

The nuclear power plant of Gösgen which powers 13% of Switzerland's energy needs.

Simulation Engineer

- End to end simulation of the irradiation cycles of fuel assemblies to ultimately compute the decay heat of those fuel assemblies using C#.
- The code was verified to other codes and concluded successfully.
- It was a project taken alongside my master's studies in nuclear engineering.

Nagra - Switzerland

Jul 2022 - Sep 2022

Nagra stands for "National Cooperative for the Disposal of Radioactive Waste".

Internship

- Implementation of a reverse Monte Carlo optimization algorithm inspired by the simulated annealing algorithm.
- Characterisation of the activation profile of Control Rods.
- Data management in recognizing functional and non-functional detectors for the Grimsel research site for geology re-search.

Mount-Lebanon Hospital - Lebanon

Jun 2020 - Dec 2020

Internship

- Fully designed application to compute the thickness of walls around LINAC rooms to ensure radiation protection.
- User-friendly application in C# with Visual Studio using the Windows Presentation Foundation (WPF) UI framework.

Internship

- Quality Control Engineer.
- Supervised the quality of service and maintenance.

ACADEMIC PROJECTS

Publications - Google Scholar profile**Oct 2023 - Current**

- **3D Reactor Kinetics:** Demonstrated the precise measurement of the prompt neutron decay constant in a zero-power reactor using "SAFFRON," a novel 150-node 3D detector system.
- **Monte Carlo Development:** Engineered and validated a new event-tracking feature within the OpenMC simulation code, specifically optimized for neutron spectroscopy and scatter camera systems.
- **Xenon-135 Observation:** Pioneered advanced gamma-ray spectroscopy methodologies aimed at directly observing and quantifying Xenon-135 poisoning dynamics in real-time.
- **Neutron Imaging:** Designed and simulated a new, high-precision neutron imaging station to expand the experimental capabilities of the CROCUS reactor facility.

Master Thesis: Temporal and spatial correlation in the CROCUS reactor with the SAFFRON 3D detector array**Feb 2023 - Sep 2023**

- Investigated the effect of neutron clustering in the CROCUS research reactor at EPFL.
- Extracted kinetics parameters of the CROCUS reactor using neutron noise techniques with Fourier transform algorithms.
- Heavy data processing due to using a novel detector named SAFFRON, composed of 150 detectors array.
- Coupled and tuned Monte Carlo codes aiming to compare experimental and simulation results.
- Based on this project, a drafted journal article and a published conference article were obtained.

EXTRA-CURRICULAR ACTIVITIES

Solidarité Liban-Suisse**Sep 2017 - Jan 2023****Volunteer**

- Procured medications from abroad to address shortages for individuals in need in Lebanon.
- Organized and participated in a gift donation initiative for orphans in Lebanon.

Association of Energy Engineers**Feb 2021 - Sep 2021****President of Student Club**

- Organized meetings and discussions focused on realistic energy solutions in Lebanon.
- Facilitated discussions on macro-scale energy strategies aimed at achieving low-carbon energy.

AWARDS

Best student paper award PHYSOR 2024

- 1st Prize for the Best Student Paper Award at the International Conference on the Physics of Reactors (PHYSOR 2024) held in San Francisco.

Dean's Award for Excellence in Teaching 2023-2024 - EPFL

- Prize for recognition of my teaching contributions at EPFL.

ADDITIONAL

- Technical Skills: C, C++, C#, Python, Matlab, Microsoft Office, Linux.
- Language Skills: English (Business Professional), French (Business Professional), Arabic (Native), German (Basic interaction)
- Hobbies: Philosophy and Ethics, Finance, Football, and Piano.